

MIT MORADABAD

Bank Management System

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Introduction

- Bank Management System is a software for organizing and managing the banking tasks digitally.
- It is an information system which is mainly focused on the operations like adding new customers in the bank, deposit , withdraw , transfer money ,statement of the customers etc.

Objectives

- Manage customer accounts (create ,update , delete)
- Perform transactions like deposits ,withdrawals , balance checks
- Maintain a history of transactions
- Ensure data security and integrity

Features of the System

- Account Management (Create View ,Update , Delete accounts)
- Transaction Management (deposit , withdraw , transfer)
- Balance inquiry
- Transaction history /Statement generation
- Secure login for user/admin

Database Design

- **Entities**:
  - **User**: Stores customer information (username, email, password).
  - **Account**: Stores account details (balance, account number).
  - **Transaction**: Logs transactions (deposit, withdrawal, transfer)..

Technology Used

PYTHON

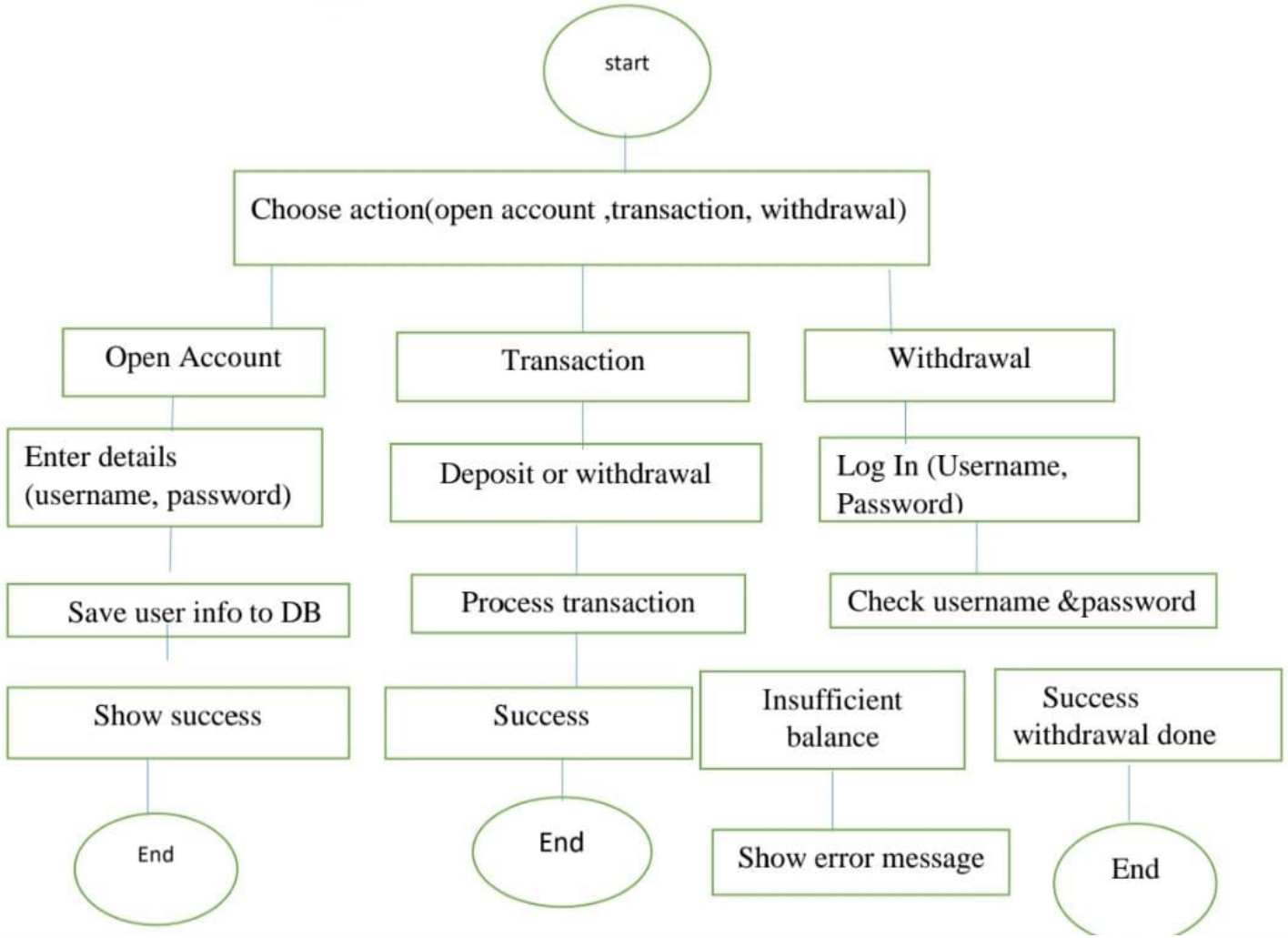
- Backend programming language.
- Easy to learn and powerful.



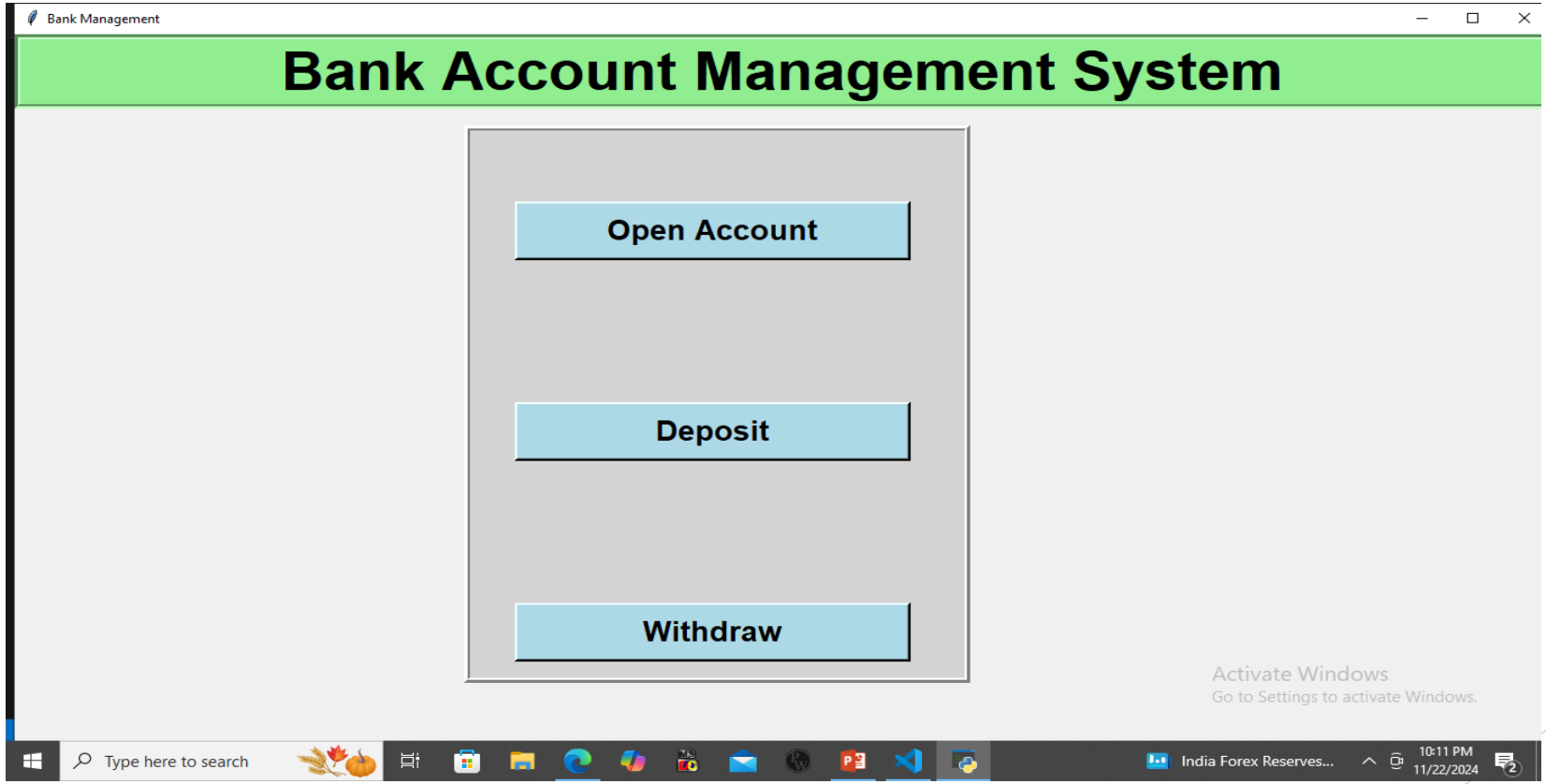
MYSQL

- Relational database management system.
- Stores user data, transactions, and account det

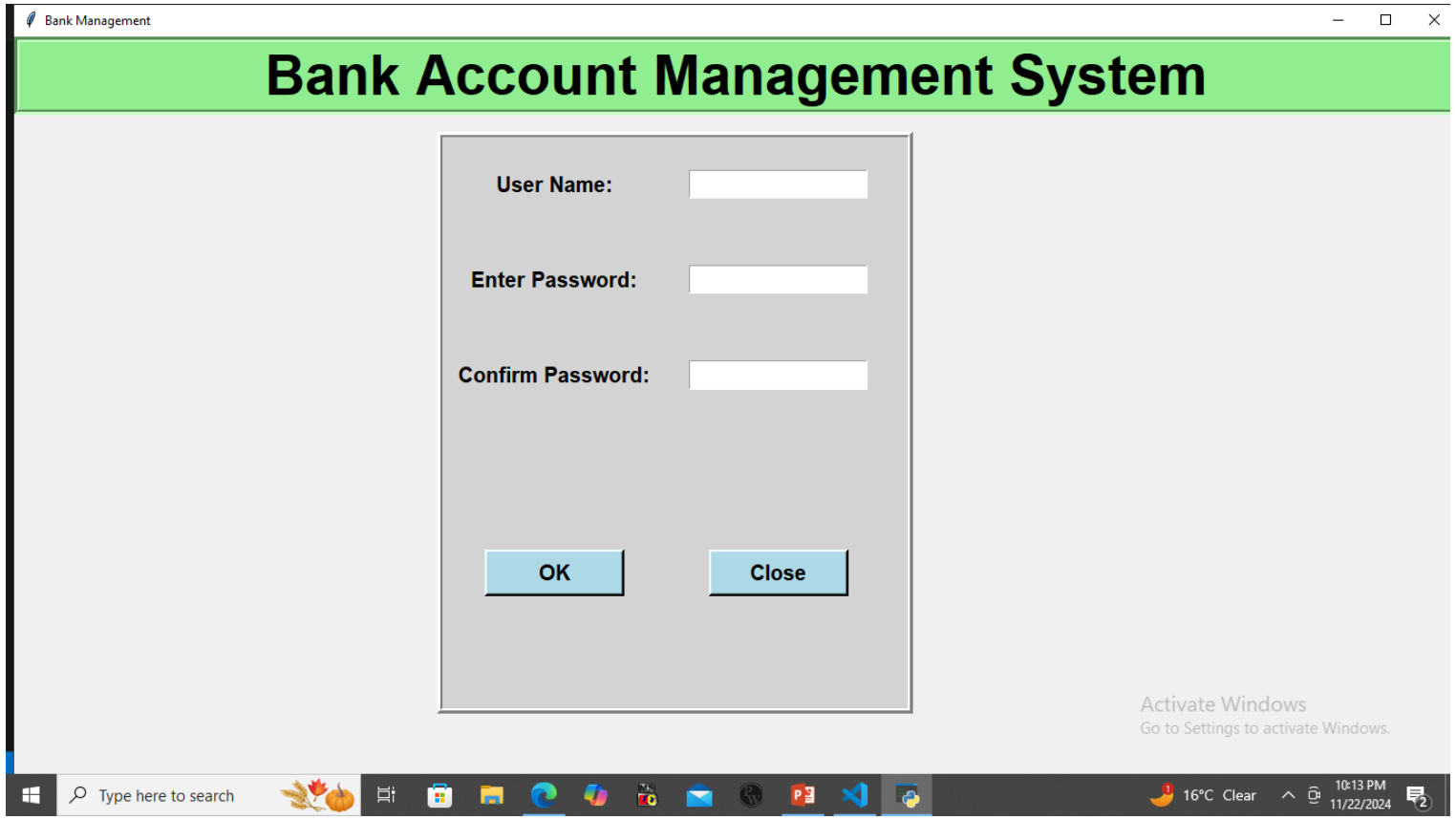
FLOW CHART


start
Choose action(open account ,transaction, withdrawal)
Open Account
Transaction
Withdrawal
Enter details (username, password)
Deposit or withdrawal
Log In (Username, Password)
Save user info to DB
Process transaction
Check username &password
Show success
Success
Insufficient balance
Success withdrawal done
End
End
Show error message
End

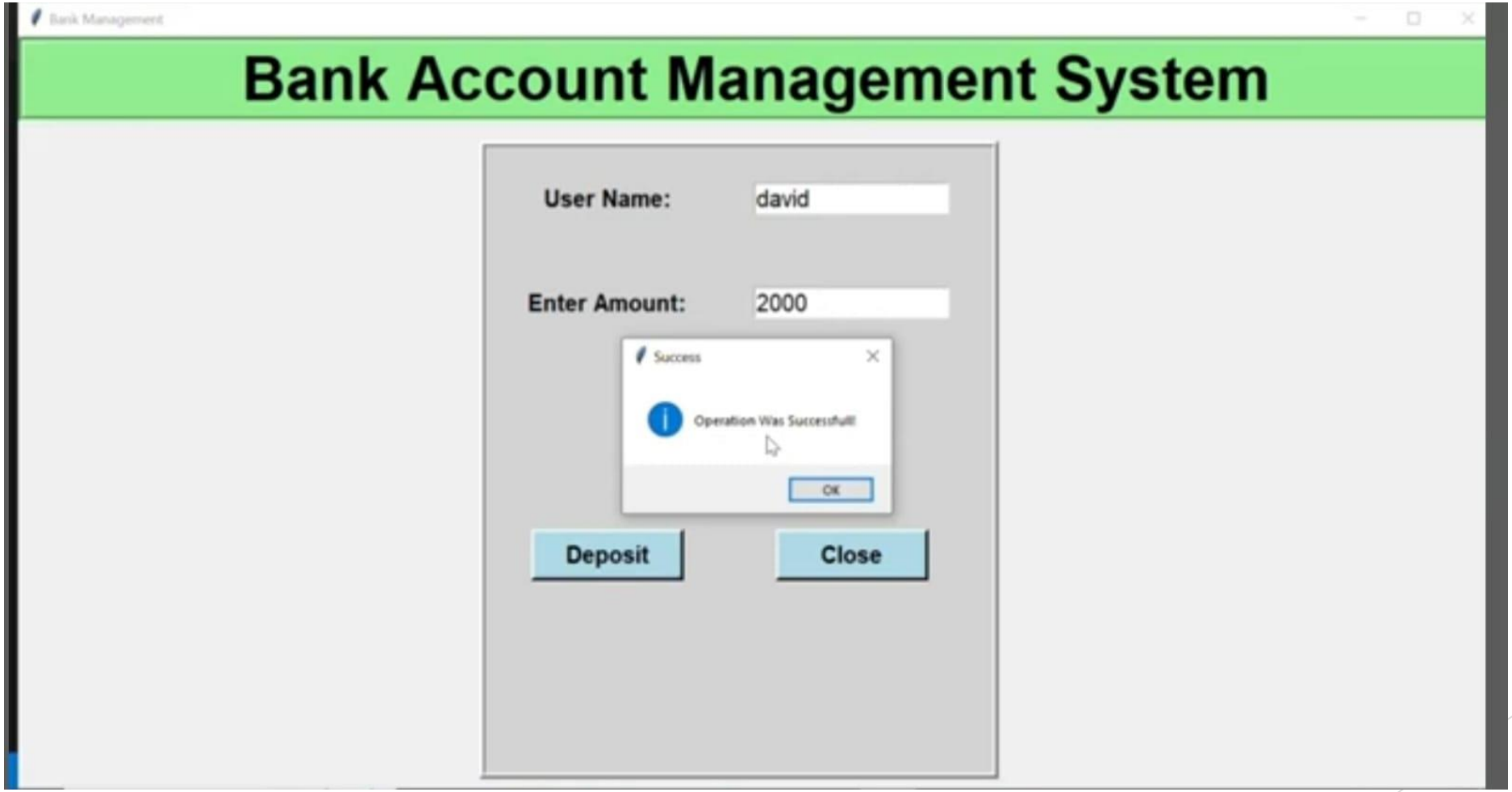
Implementation



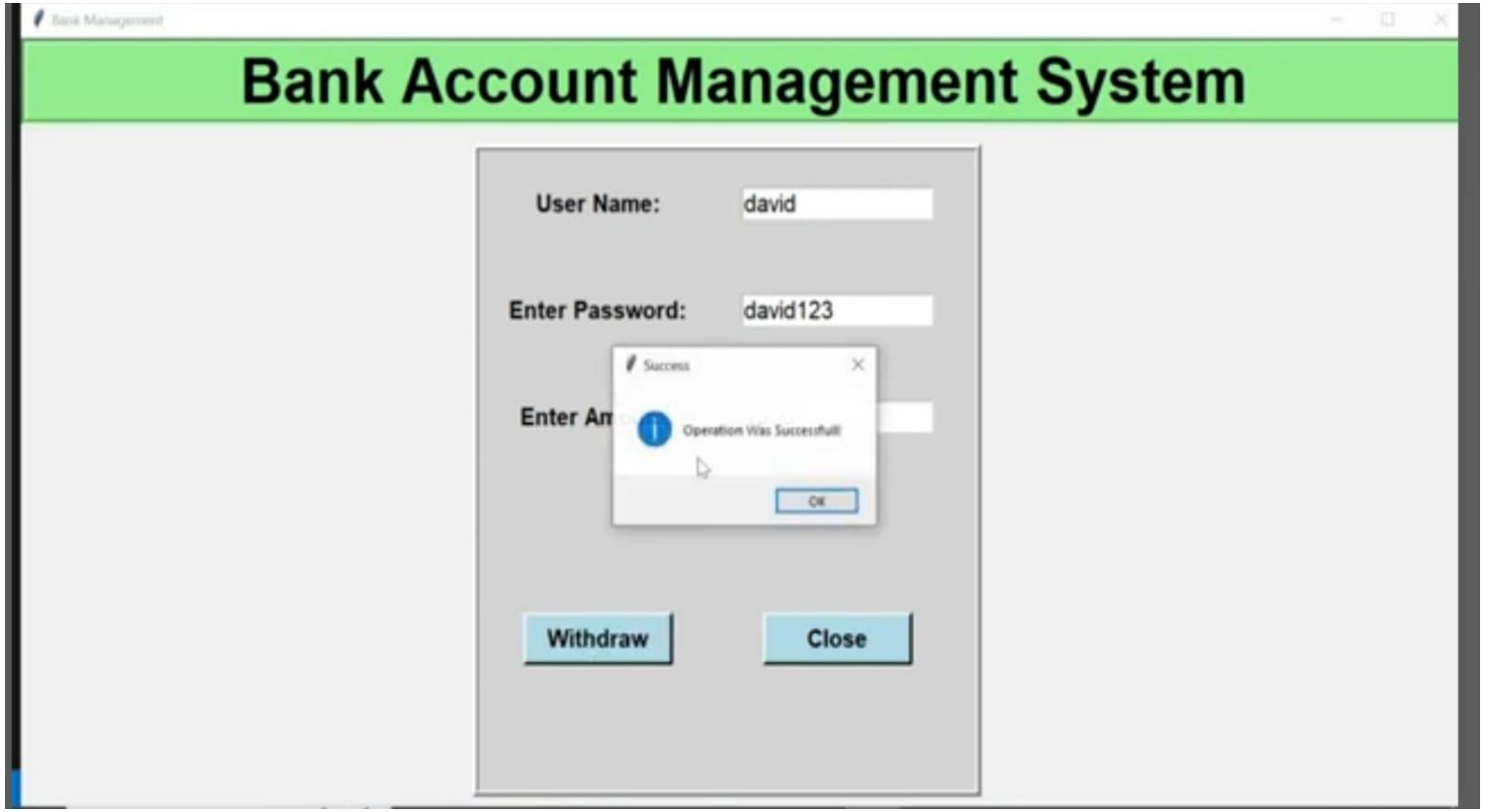
Open Account



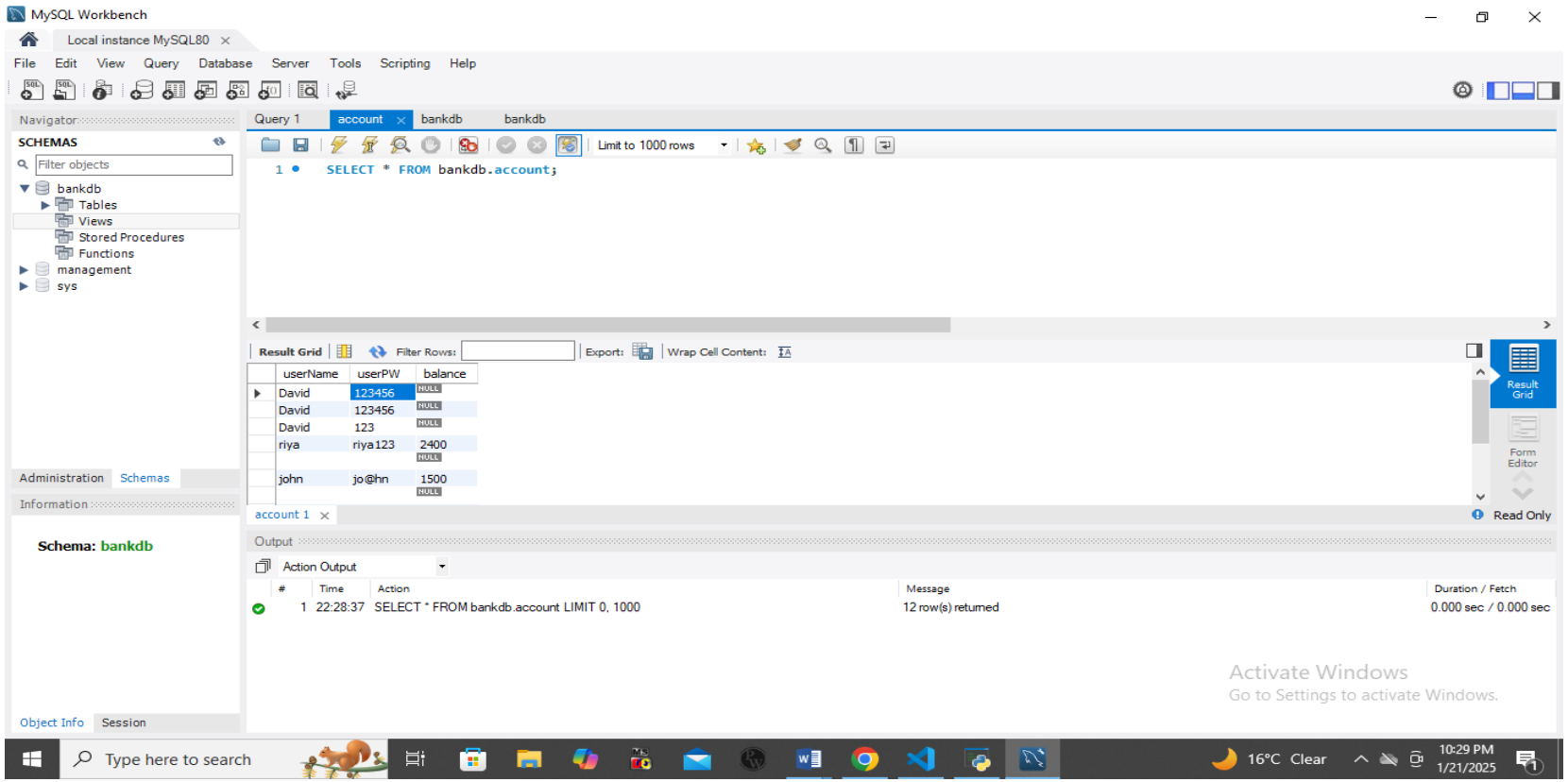
Deposit


Bank Management
Bank Account Management System
User Name:
david
Enter Amount:
2000
Success
Operation Was Successfull!
OK
Deposit
Close

Withdraw


Bank Account Management System
User Name:
david
Enter Password:
david123
Success
Operation Was Successfull!
OK
Withdraw
Close

Data store


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
bankdb
Tables
Views
Stored Procedures
Functions
management
sys
Administration Schemas
Information
Schema: bankdb
Object Info Session
Query 1 account bankdb bankdb
Limit to 1000 rows
1 SELECT * FROM bankdb.account;
Result Grid Filter Rows: Export: Wrap Cell Content:
userName userPW balance
David 123456 NULL
David 123456 NULL
David 123 NULL
riya riya123 2400 NULL
john jo@hn 1500 NULL
account 1
Read Only
Output
Action Output
Time Action Message Duration / Fetch
1 22:28:37 SELECT * FROM bankdb.account LIMIT 0, 1000 12 row(s) returned 0.000 sec / 0.000 sec
Activate Windows
Go to Settings to activate Windows.
Type here to search
16°C Clear
10:29 PM
1/21/2025

Challenges

- Discuss potential challenges in implementing and managing the system:
- Data security threats
- System downtime
- Integration with other financial services

Benefits of a Bank management System

- Time-saving and automation of banking process
- Better customer service with personalized solutions
- Improved accuracy and reduced human errors
- Enhanced security and fraud prevention

Conclusion

The **Bank Management System** provides a secure, efficient, and user-friendly platform for banking operations .The system performs well for small to medium-scale applications .With further optimization and mobile support, the system can scale for larger user bases and more complex operations.

Reference

- https://www.djangoproject.com/
- https://www.mysql.com/

THANK YOU